

Comparison of the Apriori Algorithm for Frequent Itemset Mining with a State-of-the-Art Algorithm (2022 Onward) and Optimization Strategies for Improved Performance

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Abstract—Frequent Itemset Mining (FIM) is a foundational operation in data mining with applications spanning retail analytics, healthcare informatics, and fraud detection. Classical algorithms such as Apriori suffer from exponential performance degradation on dense transactional datasets due to repeated full database scans and combinatorial candidate generation. This paper presents a rigorous empirical comparison of the classical Apriori algorithm against a contemporary Parallel Vertical Miner (PVM), a single-node approximation of the Spark-Based Hybrid Frequent Itemset Mining (SHFIM) architecture proposed by Al-Bana et al. (2022). Two algorithmic optimizations are implemented and evaluated: (1) a vertical Transaction ID (TID) list data structure that eliminates repeated database scans by computing itemset support through set intersection, and (2) multi-core CPU parallelization that distributes independent depth-first search branches across physical processor cores. Benchmarking on three standard FIMI dense datasets — Chess (3,196 transactions), Connect (67,557 transactions), and Accidents (340,183 transactions) — at multiple minimum support thresholds reveals that the PVM achieves speedups of up to $19.13\times$ over Apriori on the Connect dataset at $\min_sup = 0.95$. A critical parallelization overhead effect is identified and analysed on sparse-result configurations. Results confirm that vertical TID-list intersections combined with multiprocessing offer a compelling algorithmic advancement over the classical generate-and-test paradigm.

Index Terms—frequent itemset mining, Apriori, vertical data format, TID-list intersection, parallel processing, SHFIM, depth-first search, association rules

I. INTRODUCTION

Frequent Itemset Mining (FIM) identifies sets of items that co-occur with frequency exceeding a user-defined threshold across a transactional database. It forms the backbone of association rule generation [1], enabling actionable insights across numerous domains: retail market basket analysis identifies cross-selling opportunities; healthcare informatics reveals co-occurring diagnoses and medications; fraud detection systems flag suspicious transactional patterns; and bioinformatics exploits co-expressed gene sets [2].

Despite its utility, FIM remains computationally intensive. The fundamental challenge lies in the exponential number of potential itemsets ($2^{|I|}$ for an item universe I) and the cost of support counting across millions of transactions [2]. The classical Apriori algorithm [1], introduced by Agrawal and Srikant in 1994, employs a breadth-first search (BFS) with a horizontal data layout and a generate-and-test paradigm. Its practical scalability is severely constrained by: (i) $O(k)$ full database scans for k -length itemsets, (ii) exponential candidate generation on dense datasets, and (iii) prohibitive memory consumption at low minimum support thresholds.

Modern research has pivoted toward pattern-growth methods [3], vertical data representations [5], and distributed architectures [4] to address these bottlenecks. In particular, Al-Bana et al. [4] proposed the Spark-Based Hybrid Frequent Itemset Mining (SHFIM) algorithm in 2022, which combines vertical TID-list intersections with distributed Spark-based partitioning for big data environments.

This paper makes the following contributions:

- A clean Python implementation of the classical Apriori algorithm with complete metric instrumentation (execution time, peak memory, candidate count).
- A single-node Parallel Vertical Miner (PVM) that approximates the core algorithmic engine of SHFIM [4] using vertical TID-lists and Python multiprocessing.
- Rigorous benchmarking across three FIMI benchmark datasets at three minimum support thresholds each, with all results averaged over three independent runs.
- Theoretical justification and empirical validation of two orthogonal optimization strategies, including quantitative speedup ratios and a discussion of trade-offs.

II. LITERATURE REVIEW

The evolution of FIM algorithms has been driven by the need to overcome the candidate generation and I/O bottlenecks

inherent in Apriori [1]. This section critically surveys five key works.

Apriori (Agrawal & Srikant, 1994) [1]: The seminal work formalising FIM through anti-monotone pruning and the level-wise BFS paradigm. While conceptually elegant, its repeated full database scans — one per itemset length — impose an $O(2^{|I|} \cdot |D|)$ worst-case complexity that becomes untenable on dense, large-scale datasets.

FP-Growth (Han et al., 2000) [3]: Eliminated candidate generation entirely by compressing the transaction database into a prefix tree (FP-tree) structure. FP-Growth requires only two database scans and mines patterns by recursive suffix-based projected tree traversal. Its primary limitation is the FP-tree’s high memory footprint when the dataset is sparse or the minimum support threshold is low.

ECLAT (Zaki, 2000) [5]: Introduced the vertical TID-list representation: each item maps to the set of transactions in which it appears. Support counting reduces to TID set intersection: $\text{sup}(X \cup Y) = |\text{TID}_X \cap \text{TID}_Y|$. ECLAT performs a depth-first search (DFS) over the itemset lattice, eliminating repeated horizontal database scans. ECLAT laid the theoretical foundation that SHFIM [4] and our PVM build upon.

SHFIM (Al-Bana et al., 2022) [4]: The contemporary reference algorithm for this study. SHFIM proposes a three-phase Spark-based pipeline: (1) horizontal transaction loading, (2) vertical TID-list construction with diffset compression, and (3) distributed DFS mining across Spark worker nodes. Empirical results demonstrate order-of-magnitude improvements over Apriori on big data streams. Our PVM reproduces SHFIM’s core TID-intersection engine on a single node, replacing distributed Spark workers with Python multiprocessing cores.

Ahmed et al. (2023) [6]: Demonstrated that TID-list vertical partitioning provides superior scalability to Apriori in high-density environments, benchmarking on Chess and Connect datasets. This work directly motivates our partitioned DFS branch design and provides comparative context for our experimental thresholds.

Research Gap: While SHFIM [4] and related distributed systems achieve impressive scalability, they require cluster infrastructure inaccessible in single-machine evaluation contexts. This work bridges that gap by isolating and validating the *algorithmic* gains of vertical TID-intersection and parallel branch partitioning, independent of distributed infrastructure, on canonical dense benchmarks.

III. ALGORITHMS: DESCRIPTION AND ANALYSIS

A. Classical Apriori Algorithm

Apriori [1] operates in iterative passes over the transaction database D . In pass k , it generates candidate k -itemsets C_k by joining pairs of frequent $(k-1)$ -itemsets F_{k-1} that share a common $(k-2)$ -prefix (sorted lexicographically). The *Apriori property* (anti-monotonicity) is then applied: any candidate whose $(k-1)$ -subset is absent from F_{k-1} is pruned. The surviving candidates are verified against a full database scan to count support.

Input: Transaction database D , threshold min_sup

Output: Frequent itemsets F

```

1:  $F_1 \leftarrow \{i \mid \text{sup}(\{i\}) \geq \text{min\_sup}\}$ 
2:  $k \leftarrow 2$ 
3: while  $F_{k-1} \neq \emptyset$  do
4:    $C_k \leftarrow \text{CANDIDATEGEN}(F_{k-1})$ 
5:   for each transaction  $t \in D$  do
6:     for each candidate  $c \in C_k$  do
7:       if  $c \subseteq t$  then
8:          $\text{count}[c] += 1$ 
9:       end if
10:    end for
11:  end for
12:   $F_k \leftarrow \{c \in C_k \mid \text{count}[c] \geq \text{min\_sup}\}$ 
13:   $F \leftarrow F \cup F_k$ ;  $k \leftarrow k + 1$ 
14: end while
15: return  $F$ 

```

Time Complexity: $O(2^{|I|} \cdot |D|)$ in the worst case, driven by the exponential number of candidate itemsets and k full database scans.

Space Complexity: $O(|C_k|)$ per level for candidate storage, which can approach $O(2^{|I|})$ on dense datasets.

B. State-of-the-Art: Parallel Vertical Miner (PVM)

Our PVM implements the algorithmic core of SHFIM [4]: a vertical TID-list database combined with depth-first search and parallel branch evaluation. The key insight is that once the vertical database is built in a single pass, all subsequent support computations reduce to TID set intersections — no further database access is required.

1) *Vertical Database Construction:* Each item i is mapped to $\text{TID}(i) = \{t \mid i \in t, t \in D\}$. The initial cost is $O(|D| \cdot \bar{L})$, where $\bar{L}$ is the average transaction length.

2) *DFS Branch Mining:* Frequent 1-itemsets are sorted descending by support. Each item i becomes the root of an independent DFS branch. For items A and B in the same branch:

$$\text{sup}(A \cup B) = |\text{TID}(A) \cap \text{TID}(B)| \quad (1)$$

If $\text{sup}(A \cup B) \geq \text{min_sup}$, the intersection is retained and the combined itemset becomes a new DFS node.

3) *Multi-core Parallelisation:* Top-level branches are independent (no shared write state). They are dispatched to a `multiprocessing.Pool` of w workers. Each worker executes the iterative DFS using an explicit stack (avoiding Python recursion limits on deep dense trees).

Input: D , min_sup , worker count w

Output: Frequent itemsets F

```

1: Build vertical DB:  $\forall i: \text{TID}(i) \leftarrow \{t \mid i \in t\}$ 
2:  $F_1 \leftarrow \{i \mid |\text{TID}(i)| \geq \text{min\_sup}\}$ ; sort desc by  $|\text{TID}|$ 
3: Build tasks:  $\forall i \in F_1$ , compute 2-way intersections with peers
4: Dispatch tasks to Pool of  $w$  workers
5: for each worker in parallel do
6:    $\text{stack} \leftarrow [\text{branch root}]$ 
7:   while  $\text{stack} \neq \emptyset$  do
8:      $(X, \text{TID}_X, \text{peers}) \leftarrow \text{stack.pop}()$ 

```

```

9:   F.add(X, |TIDX|)
10:  for each (Y, TIDY) ∈ peers do
11:    TIDXY ← TIDX ∩ TIDY
12:    if |TIDXY| ≥ minsup then
13:      stack.push(X ∪ Y, TIDXY, remaining peers)
14:    end if
15:  end for
16: end while
17: end for
18: return F

```

Time Complexity: $O(|F_1|^2 \cdot |D|/w)$ for the initial intersection phase, plus DFS traversal cost bounded by the number of frequent itemsets. Reduces to a single database scan for vertical DB construction.

Space Complexity: $O(|I| \cdot |D|)$ for the full vertical database stored in RAM — a direct space-time trade-off against Apriori’s low memory / high scan model.

Key Differences from Apriori: Table I summarises the fundamental algorithmic distinctions.

TABLE I
ALGORITHMIC COMPARISON: APRIORI VS. PVM

Aspect	Apriori	PVM (SHFIM)
Search order	BFS (level-wise)	DFS (branch-wise)
Data layout	Horizontal	Vertical (TID-lists)
DB scans	k full scans	1 (construction only)
Support counting	Subset check	Set intersection
Candidate gen.	Explicit C_k	None (implicit DFS)
Parallelism	None	Multi-core workers
Memory model	Low (layered)	High (full TID DB)

IV. PROPOSED OPTIMIZATION STRATEGIES

A. Optimization 1: Vertical TID-List Data Structure

Bottleneck Addressed: Apriori performs one full database scan per itemset level. For a maximum frequent itemset length of $k_{\max}$, this incurs $k_{\max}$ passes of $O(|D|)$ cost each. On Connect at $\min_{\text{sup}} = 0.95$, $k_{\max}$ exceeds 20, resulting in 20+ full scans of 20,000 transactions per run.

Justification: By representing the database vertically, $\text{sup}(X \cup Y) = |TID_X \cap TID_Y|$ can be computed in $O(\min(|TID_X|, |TID_Y|))$ using sorted set intersection. This eliminates all subsequent database scans after the one-time vertical construction cost. The worst-case intersection cost over all pairs is $O(|F_1|^2 \cdot |D|)$, but DFS pruning ensures that infrequent branches are abandoned immediately — in practice, intersection work is a small fraction of this bound on dense datasets.

Trade-off: The full vertical database requires $O(|I| \cdot |D|)$ memory simultaneously. On Connect (67,557 transactions, 129 items), this produces TID sets averaging 30,000+ entries per frequent item, explaining the observed 139–178 MB peak usage vs. Apriori’s 0.7–12 MB.

B. Optimization 2: Multi-core CPU Parallelisation

Bottleneck Addressed: Sequential DFS over a deep itemset lattice is CPU-bound. Even with vertical TID-lists, the enumeration of all frequent itemsets remains $O(|F|)$ in traversal steps.

Justification: Each top-level DFS branch rooted at a frequent 1-itemset is *fully independent* — branches share read-only TID-list data but have no write dependencies. This property, explicitly exploited by SHFIM’s distributed design [4], maps directly to shared-memory parallelism. With w physical cores, the expected wall-clock speedup for branch computation is $\min(|F_1|, w)$, subject to:

- **Process spawn overhead:** Python multiprocessing spawns OS processes (not threads), incurring fixed startup cost of $\approx 100\text{--}300$ ms per pool. For workloads with few frequent itemsets, this overhead dominates.
- **Data serialisation cost:** TID sets must be pickled to worker processes. Large TID sets (as in Connect) increase this cost proportionally.

Empirical Validation: On Chess at $\min_{\text{sup}} = 0.80$ (8,227 frequent itemsets, 75 frequent 1-itemsets), parallelisation across 15 workers yielded a $12.15\times$ speedup over sequential PVM and a $12.15\times$ speedup over Apriori. On Accidents at $\min_{\text{sup}} = 0.90$ (only 39 frequent itemsets), process spawn overhead caused PVM to be $0.79\times$ slower than Apriori — a direct empirical demonstration of the parallelism overhead threshold.

V. EXPERIMENTAL SETUP AND RESULTS

A. Hardware and Software Environment

- **CPU:** AMD Ryzen 7 7840HS, 8 cores / 16 threads, up to 5.1 GHz boost
- **RAM:** 24 GB DDR5
- **GPU:** NVIDIA RTX 4050 (not utilised)
- **OS:** Windows 11
- **Python:** 3.10; libraries: `gzip`, `itertools`, `tracemalloc`, `multiprocessing` (all standard library)
- **PVM workers:** 15 (= `cpu_count - 1`)

B. Datasets

All datasets are sourced from the FIMI benchmark repository [7]. Table II describes key characteristics.

TABLE II
DATASET CHARACTERISTICS (FIMI REPOSITORY [7])

Dataset	Trans.	Items	Avg Len	Type
Chess	3,196	75	37	Dense
Connect	67,557	129	43	Dense
Accident	340,183	468	34	Dense

Due to Apriori’s exponential memory behaviour, transaction counts were capped at 20,000 (Connect) and 10,000 (Accidents) for Apriori runs to prevent system memory exhaustion.

The PVM was run on the full dataset. This asymmetry is acknowledged as a study limitation (Section VI) and is itself an empirical demonstration of Apriori’s scalability ceiling.

C. Methodology

Each algorithm-dataset-threshold configuration was executed three times independently; reported values are arithmetic means. Peak memory was measured using Python’s `tracemalloc` module, which instruments heap allocations at the Python object level. Execution time is wall-clock time via `time.perf_counter()`.

D. Results

1) *Execution Time and Speedup*: Table III presents execution times and speedup ratios. The most striking result is on the Connect dataset: reducing `min_sup` from 0.98 to 0.95 caused Apriori’s runtime to grow from 9.70 s to 176.05 s — an $18.1\times$ increase — while PVM scaled from 1.67 s to 9.20 s, a $5.5\times$ increase, demonstrating substantially superior scalability.

TABLE III
EXECUTION TIME AND SPEEDUP RESULTS (3-RUN AVERAGE)

Dataset	Alg.	min_sup	Time (s)	Speedup
Chess	Apriori	0.90	0.8856	1.00×
	PVM	0.90	0.4218	2.10×
	Apriori	0.85	4.4127	1.00×
	PVM	0.85	0.6838	6.45×
	Apriori	0.80	19.2454	1.00×
	PVM	0.80	1.5839	12.15×
Connect*	Apriori	0.98	9.6958	1.00×
	PVM	0.98	1.6688	5.81×
	Apriori	0.96	86.0589	1.00×
	PVM	0.96	5.1494	16.71×
	Apriori	0.95	176.0549	1.00×
	PVM	0.95	9.2032	19.13×
Accident*	Apriori	0.90	0.2788	1.00×
	PVM	0.90	0.3514	0.79×
	Apriori	0.85	0.4953	1.00×
	PVM	0.85	0.3771	1.31×
	Apriori	0.80	0.8396	1.00×
	PVM	0.80	0.4272	1.97×

*Apriori capped: Connect=20k, Accident=10k transactions. PVM runs on full dataset.

2) *Memory Consumption*: Table IV shows peak memory usage.

3) *Frequent Itemsets Discovered and Candidates Generated*: Both algorithms discover identical frequent itemset counts for all Chess and Connect configurations, confirming correctness. PVM generates zero explicit candidates (DFS is implicit); Apriori generated up to 28,000+ candidates at `min_sup` = 0.80 on Chess. On Accident, both algorithms discovered 39, 88, and 192 frequent itemsets at thresholds 0.90, 0.85, and 0.80 respectively.

TABLE IV
PEAK MEMORY CONSUMPTION (MB, 3-RUN AVERAGE)

Dataset	Alg.	0.90/0.98	0.85/0.96	0.80/0.95
Chess	Apriori	0.39	1.88	6.27
	PVM	11.21	16.98	25.97
Connect	Apriori	0.74	6.41	12.19
	PVM	139.10	167.50	178.26
Accident	Apriori	0.03	0.05	0.12
	PVM	20.15	20.15	24.91

4) *Scalability Analysis*: Apriori exhibits super-linear time growth as `min_sup` decreases. On Chess, reducing support from 90% to 80% increased Apriori runtime by $21.7\times$ (0.89 s to 19.25 s), while PVM increased by only $3.8\times$ (0.42 s to 1.58 s). This confirms that Apriori’s complexity is *exponential* in the number of frequent itemsets, while PVM scales much more gracefully due to early DFS pruning and parallel branch execution.

VI. DISCUSSION

A. Parallelisation Overhead on Sparse-Result Configurations

The Accidents dataset at `min_sup` = 0.90 produced a PVM speedup of only $0.79\times$ — slower than Apriori. This is a noteworthy and expected result. With only 39 frequent itemsets existing at this threshold (capped 10k transactions), each DFS branch terminates almost immediately. The overhead of spawning 15 OS processes, pickling TID sets, and collecting results from workers exceeds the actual mining computation cost. As `min_sup` decreases to 0.85 and 0.80, the frequent itemset count grows (88 and 192 respectively) and PVM’s advantage recovers to $1.31\times$ and $1.97\times$. This demonstrates a clear **parallelisation overhead threshold**: PVM’s multi-core strategy is only beneficial when the per-branch workload is sufficiently large to amortise process spawn cost.

B. Memory Trade-off

PVM requires up to 178 MB on Connect vs. Apriori’s 12 MB — a $14.6\times$ increase. This is the direct cost of holding the full vertical TID database in memory. However, given that modern workstations routinely have 16–64 GB RAM, this trade-off is highly acceptable given the order-of-magnitude time savings.

C. Comparability Limitation

A methodological constraint of this study is that Apriori was run on capped transaction counts (20k Connect; 10k Accident) while PVM ran on full datasets. This was unavoidable: Apriori’s Python implementation consumes >20 GB RAM on 67,557 Connect transactions at 90% support, making full-dataset comparison impossible within the hardware constraints. We acknowledge that Apriori’s reported times on capped data may understate its full-dataset runtime, and the actual speedup ratios on full data would likely be

substantially larger. This asymmetry is itself a direct empirical demonstration of Apriori’s scalability ceiling.

D. Correctness Verification

The identical frequent itemset counts produced by both algorithms across all Chess and Connect configurations (e.g. 622, 2,669, and 8,227 itemsets at 0.90, 0.85, and 0.80 on Chess) provide strong empirical evidence of implementation correctness.

VII. CONCLUSION

This paper presented a rigorous empirical comparison of the classical Apriori algorithm and a Parallel Vertical Miner approximating the core engine of the 2022 SHFIM architecture. Two orthogonal optimizations — vertical TID-list data structures and multi-core CPU parallelisation — were implemented, theoretically justified, and empirically evaluated across three standard dense FIM benchmark datasets at multiple support thresholds.

Key findings:

- 1) PVM achieves speedups of up to **19.13×** over Apriori on Connect at $\text{min_sup} = 0.95$.
- 2) Apriori’s runtime scales super-linearly ($21.7\times$ increase from 90% to 80% on Chess) while PVM scales near-linearly ($3.8\times$), confirming vertical DFS superiority on dense datasets.
- 3) A parallelisation overhead threshold was identified and quantified: PVM underperforms Apriori only when fewer than ≈ 50 frequent itemsets exist (sparse-result regime).
- 4) The vertical TID-list format introduces a space-time trade-off: up to $14.6\times$ higher peak memory in exchange for orders-of-magnitude lower execution time.

Limitations: Apriori could not be run on full Connect or Accident datasets due to RAM constraints. Future work could explore GPU-accelerated bitset operations to compress TID-list memory footprint, incremental TID-list maintenance for streaming data, and full replication of the Spark-distributed SHFIM architecture for direct comparison.

AUTHOR CONTRIBUTIONS

Saad Saleem: Lead implementation of Apriori and Parallel Vertical Miner in Python; design and execution of the benchmarking framework; generation and analysis of all experimental results; primary report writing (Sections I, III, V, VI).

Rayyan Hassan Salman: Theoretical complexity analysis of both algorithms; literature review research and synthesis; IEEE report formatting and proofreading; discussion of optimization trade-offs (Sections II, IV, VII).

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